

Research and Experiment Report

1. Problem Statement

Goal: The primary objective of this project is to empirically evaluate the capability of a large language model (LLM), specifically GPT-4o-mini, to accurately simulate the conversational responses, with a particular focus on syntax and semantics, of principal characters from the television series *Friends*. This simulation is designed to occur without reliance on direct memorization from the training data (i.e., using the provided dialogue dataset as a reference, not a source of direct replication). The choice of the *Friends* dataset is motivated by the desire to understand the potential of AI chatbots to serve as believable companion or "friend" figures in a conversational context.

Inputs and Outputs:

- **Input:** A precisely engineered prompt template that instructs the LLM to adopt the persona of a specific *Friends* character. For example: "You are [Character Name] from *Friends*. Respond naturally as you would in the show, continuing the conversation below." The input also includes the preceding dialogue lines to provide conversational context.
- **Output:** A generated text response from the LLM that is intended to faithfully emulate the linguistic style, emotional tone, and semantic content characteristic of the target character.

Connection to Course Requirements: This investigation directly addresses and explores the advanced limits of LLMs in areas such as nuanced character simulation, emotional reasoning, and complex trait generalization - key themes in contemporary artificial intelligence research. The project aligns with core course objectives by requiring the application of advanced prompt engineering techniques, the implementation of automated prompting via structured templates, and the development of custom empirical evaluation methodologies. Specifically, the use of metrics such as ROUGE, BERTScore, cosine similarity, and a novel approach to measuring memorization will be employed to quantitatively assess and measure the performance of the LLM.

2. Dataset

Dataset Name and Source:

- **Name:** *Friends* TV Show Transcripts
- **Source:** Cornell Convokit Dataset - *Friends* Corpus

The dataset comprises comprehensive dialogue transcripts, encompassing over 60,000 individual dialogue lines across all 10 seasons of the *Friends* series. The corpus is curated and specifically structured for in-depth conversational and linguistic analysis.

Attributes:

- **season_id:** The season number of the episode.
- **episode_id:** The episode number within the corresponding season.
- **scene_id:** The scene number within the episode.
- **utterance_id:** The sequential order of the utterance within a given scene.
- **speaker:** The character who delivered the line (e.g., Ross, Rachel, Chandler).
- **tokens:** An approximate count of word units (or tokenized words).
- **transcript:** The raw text of the actual dialogue line.

Dataset Statistics:

- **Total Examples (Initial):** 67,373
- **Total Examples (After Cleaning and Filtering):** 51,297 (Filtered for completeness and limited to main characters)
- **Data Structure:** 7 columns, all of the object data type.
- **Duplicate Rows:** None detected.
- **Missing Values:** Only found in the transcript column (6,063 missing values, subsequently removed).
- **Input Characteristics:**
 - **tokens:** Nested list of tokenized words per utterance.
 - **Average Length:** Approximately 5–15 tokens per utterance.
- **Output Characteristics:**
 - **transcript:** Raw text of the spoken line.
 - **Variable length,** ranging from a full sentence to a brief phrase.
- **Speakers:** Filtered and limited to the six main characters from *Friends* (Ross, Rachel, Monica, Chandler, Joey, Phoebe).
- **Speaker Distribution:** The utterance count is balanced, ranging from approximately 7,500 to 9,300 utterances per main character.

Dataset Creation and Preprocessing Steps:

- All lines spoken by non-core, recurring, or minor characters were systematically removed.
- The dataset was filtered to include only lines possessing complete speaker and

transcript data.

- For the purpose of evaluation metric calculation (specifically the memorization metric), further refinements were applied:
 - Entire conversations involving any characters other than the six main *Friends* characters were excluded.
 - Rows with empty tokens columns were retained, as they often represented natural conversational breaks or the end of a dialogue turn.
 - Conversations involving fewer than three unique speakers were removed, as they were deemed insufficient for providing adequate conversational context for the LLM to respond as the third speaker.

3. Prompt Methodology

Prompt Template:

The core instruction provided to the LLM is defined by the following template:

"You are {test['speaker']} from Friends. Respond as you would in the show, continuing the conversation below."

This concise prompt was intentionally designed to minimize external influence, thereby isolating and evaluating the LLM's inherent capacity to naturally simulate character-specific syntax, semantics, and tone.

Sample Input/Output Example:

- **Context:**
 - Monica Geller: There's nothing to tell! He's just some guy I work with!
 - Joey Tribbiani: C'mon, you're going out with the guy! There's gotta be something wrong with him!
- **Actual (Chandler Bing Reference Dialogue):** All right Joey, be nice. So does he have a hump? A hump and a hairpiece?
- **GPT Reply (Generated Output):** "Yeah, like he probably thinks 'a moment of silence' is when you pause for applause at a bad magic show. I mean, he can't just be a normal guy. That's like a unicorn that doesn't sparkle! "

Sampling Parameters:

Parameter	Value
Model	gpt-4o-mini ▾
Temperature	1.0 ▾
Top-p	1.0 ▾
Max Tokens	150 ▾
Stop Tokens	Not used ▾

- **Temperature and Top-p:** These values were maintained at their maximum/default settings (1.0) to encourage maximum creativity and variability in the generated responses, which is critical for assessing the model's ability to produce novel dialogue rather than direct copies.
- **Max Tokens:** A limit of 150 tokens was imposed to ensure the generated outputs remained concise, reflecting the typical, short-form conversational style of sitcom dialogue.

API Call Description:

The OpenAI Python SDK was utilized to interface with the gpt-4o-mini model. The structure of the API call was standardized to include:

- A **system message** to formally define the character role and the simulation task.
- A **user message** containing the two preceding lines of dialogue to establish the conversational context.
- The generated output was subsequently retrieved and systematically stored alongside the calculated evaluation scores (including cosine similarity, ROUGE, and BERTScore).

4. Evaluation Approach

Metrics Utilized:

- **Memorization (Cosine Similarity):** My project aims for the LLM to demonstrate genuine intelligence by generating novel outputs that are stylistically and semantically similar but not identical to the dataset. Memorization is evaluated by constructing a test set of conversations limited to the main *Friends* characters and involving more than three speakers to ensure robust context. The context (the dialogue of the first two characters)

is provided to the LLM, which is instructed to respond as the third character. The cosine similarity is then calculated between the generated GPT response and the actual third character's dialogue from the reference transcript. A low score indicates successful generalization rather than direct memorization.

- **ROUGE and BERT Scores (Syntax and Semantics):** These metrics were incorporated based on mentor recommendation, recognizing their foundational importance in evaluating natural language generation, particularly concerning syntax and semantics (a core concept from the introductory NLP course). Achieving similarity in both syntax and semantics is considered the definitive criteria for successful character simulation.
 - **BERTScore:** Calculates the semantic similarity between the generated GPT response and the actual dialogue using contextual embeddings, reflecting how similar the *meaning* is.
 - **ROUGE Score:** Measures the n-gram overlap between the GPT response and the actual dialogue, assessing surface-level syntactic similarity. I calculated:
 - **ROUGE-1 (Unigrams):** Overlap of individual words.
 - **ROUGE-2 (Bigrams):** Overlap of contiguous two-word sequences.
 - **ROUGE-L (Longest Common Subsequence):** Based on the longest sequence of words common to both the generated and reference texts, regardless of gaps.

Evaluation Process:

The evaluation was fully automated using custom code, leveraging libraries for calculating ROUGE and BERT scores, as well as for conducting the cosine similarity analysis. The use of cosine similarity was also recommended by the mentor as a suitable empirical measure of vector similarity between the generated and reference text embeddings.

Strengths and Weaknesses: Initially, I encountered difficulties in formalizing appropriate, quantitative metrics for the open-ended problem of character simulation. However, following targeted feedback from the mentor, the selected metrics - BERTScore for semantics, ROUGE for syntax, and cosine similarity for non-memorization - are now considered to comprehensively capture the core goals outlined in the problem statement, providing a robust empirical foundation for the analysis.

5. Results

Summary of Key Metrics:

METRIC	AVERAGE SCORE
Memorization (Cosine Similarity)	0.2190
ROUGE-1 F1	0.08424
BERTScore F1	0.83045

Discussion and Interpretation:

While the overall utility of the generated outputs is mixed, the results indicate that the LLM possesses strong semantic understanding but would benefit from further parameter tuning or training focused on language syntax fidelity. Due to the substantial size of the cleaned dataset, a pragmatic decision was made to conduct the comprehensive evaluation on a representative sample of ten (10) entries. The following interpretations are based on this sample:

- **Low Cosine Similarity (0.2190):** This critically low score indicates that the LLM successfully avoided copying or memorizing direct responses from the training data. This result strongly demonstrates a genuine capability for **generalization**, as the model produced novel and varied outputs, a key objective of the project.
- **Low ROUGE-1 Scores (0.08424):** The low n-gram overlap with the actual show transcripts suggests limited surface-level similarity. The model is not relying on the same words or common phrases as the reference, indicating a low degree of **syntactic mimicry**. This implies the model is generating highly creative or paraphrased outputs rather than direct replications, which accounts for the low score even if the meaning is preserved.
- **High BERTScore F1 (0.83045):** The high BERTScore reflects a strong degree of **semantic similarity** between the generated and reference dialogues, even when the exact word choices differ. This is the most significant positive finding, implying that the generated responses successfully stayed on-topic, captured the intended meaning, and maintained the emotional and conversational flow required for the character, even if they were not verbatim copies.